# 334 A simple transfer method for Example-Based Machine Translation

Tantely Andriamanankasina† Kenji Araki‡ Yoshikazu Miyanaga† Koji Tochinai†
†Hokkaido University ‡Hokkai-Gakuen University

## 1. Introduction

We [1] has presented a new matching method for the Example-Based Machine Translation. During the search for the best similar sentence, it selects the shortest sentence having the maximum number of two consecutive words equivalences. It differs from the "length first" method proposed by Niremburg [2] , since the search of word equivalence starts from left to right in a sentence, and from the DP-Matching method proposed by Kitano [3], since it compares words two by two, not one by one. This paper presents a transfer operation for this method. Other than the best outcome, it uses a set of segments which match exactly any possible segment of the input sentence. These are used to translate any word or any segment of words. The transfer process proposed in this paper is not based on the matching graph presented by Watanabe [4]. It simply observes word positions and attempts to replace, modify or insert words in the target sentence, which is the translation of the best similar sentence. The system is implemented in French-Japanese machine translation.

## 2. The transfer process

The matching process gives the sentence which matches the best the input sentence, from the translation examples. In a sentence, the system first attempts to join together consecutive words which are considered to be dependent. This join operation is applied to the input sentence, the best outcome and its Japanese translation. For example, as far as French is concerned, a determinant or a preposition is combined with the word following it. In the case of the Japanese, a particle is combined with its predecessor. It is supposed that the translation of these grouped segment forms also a group in the target language.

The system translates each segment, using a set of translation examples previously selected. These examples contain segments which match exactly segments of input sentence. For every word in the input sentence, the sentence having the longest segment which match the segment starting from that word, is selected. If exact match expires, comparison on part-of-speech tag from the next word will make the difference. Picking up the translations of segments of the input sentence becomes straightforward.

The last step of the process is an operation of insertion, deletion and replacement. Using the Japanese sentence of the example translation, the target sentence is obtained by deleting segment having no correspondent, and replacing segment where its French correspondent is different from the segment of the input sentence. The most difficult task is to find where a segment which has no correspondent should be inserted. The next best matches will give information about it, if the segment has a correspondent there. If that is not the case, it is placed next to the translation of the segment next to its translation in the source sentence.

## 3. Examples

This paragraph gives an illustration of the method. The input sentence is presented in [Table 1] with 2 best matches.

Table 1. Input sentence and best matches

| Input sentence | *quand est-ce que vous partez pour Paris ?* |
|---|---|
| 1st best match | *est-ce que Jean est à la maison ?*<br>ジャンさんは家にいますか? |
| 2nd best match | *où allez-vous ce soir ?*<br>今晩どこへ行きますか? |

- Step 1 : Joining words which are considered to be dependent between themselves :

| Input sentence | *quand/ est-ce que/ vous/ partez/ pour Paris/ ?* |
|---|---|
| 1st best match | *est-ce que/ Jean/ est/ à la maison/ ?*<br>ジャンさんは / 家に / います / か / ? |

- Step 2 : Searching the meaning of each segment :

| Sentence selected for the segment *quand* | *quand sera la prochaine fête ?*<br>次の祭りはいつですか? |
|---|---|
| Deduction of the meaning | *quand* → いつ |
| In the same way, translations of all segments are obtained | *quand*/いつ, *est-ce que*/null, *vous*/あなたは, *partez*/行きます, *pour Paris*/パリへ |

- Last step : Construction of the target sentence

| Best matching sentence : | ジャンさんは家にいますか? |
|---|---|
| By similarity, we have : | あなたはパリへ行きますか? |
| Insertion of the segmentいつ (by similarity with どこへin the 2nd best match) | あなたはパリへいつ行きますか? |

## 4. Conclusion

In this paper, we have presented with illustration a simple transfer method for Example-Based Machine Translation. Possible improvement of the method and future directions of the research depend on the result of this study.

## 5. References

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